

Lakshya Bajaj

lakshyabajaj14@gmail.com | +1 (773) 531-0057 | linkedin.com/in/lakshyabajaj | West Lafayette, IN

Education

Purdue University, *West Lafayette, IN*

August 2024 – May 2028

B.S. in Robotics Engineering Technology; Minor in Business Economics (Dean's List x 3)

GPA: 3.78/4.00

Relevant courses: Industrial Controls, Analog Signal Processing, Industrial Robot Programming, Graphic Communication & Spatial Analysis, Design Thinking in Technology

Projects and Research

Undergraduate Research, Mechanisms and Robotic Systems Lab, *West Lafayette, IN*

January 2026 – Present

- Designed schematics and a custom PCB to interface an 8x8 piezoresistive tactile sensor array with an Arduino Nano, enabling structured data acquisition for downstream force estimation
- Developed signal isolation and multiplexing strategies to ensure single-node readout, reducing cross-talk and improving measurement fidelity across the sensor matrix
- Implemented circuit-level noise reduction and signal conditioning techniques to improve sensor reliability and consistency during tactile data capture
- Used KiCad for schematic capture and PCB layout, translating tactile sensing requirements into manufacturable hardware for a learning-based 3D force estimation pipeline

Robotics Engineer, Humanoid Robot Club, *West Lafayette, IN*

August 2024 – May 2025

- Earned the AMD Most Innovative Design of the Year award for contributing to the innovative "chicken leg" foot design
- Researched and designed electrical systems for humanoid legs, focusing on motor control, IMUs, and microcontrollers

Embedded Systems Engineer, Fretboard Trainer, *West Lafayette, IN*

January 2025 – May 2025

- Secured 3rd place out of 130 teams in Purdue's Design and Innovation Challenge
- Conducted ethnographic research on music learning methods via surveys, interviews, observations, and literature reviews
- Engineered an engaging, portable, and affordable device to help aspiring musicians learn chords

Work Experience

Production Engineering Intern, NuTaste Food and Drink Labs, *Gurugram, India*

June 2025 – August 2025

- Designed and implemented a shift log sheet to track batch status, reducing handover time and improving process transparency; oversaw staff training on usage
- Applied lean manufacturing principles, including Kaizen and 5S methodology, to improve operational efficiency by 4%
- Collected and analyzed cycle time and process time data for labeling and capping, key bottlenecks in production
- Conducted a complete Value Stream Mapping exercise for one batch to identify inefficiencies and streamline operations
- Performed a cost-benefit analysis comparing square and round bottles, evaluating packaging density, truck space utilization, and equipment compatibility

Leadership Positions

Team Leader International, Purdue Orientation Programs, *West Lafayette, IN*

March 2025 – September 2025

- Mentored and guided 35 first-year students during Purdue's nationally recognized freshman orientation program
- Facilitated icebreakers, discussions, and team-building activities to ease cultural adjustment and build community
- Oversaw integration of international and domestic students, fostering cross-cultural connections and peer support
- Completed 30+ hours of leadership and cultural sensitivity training; contributed 90+ hours during orientation week

Skills

Robotics, Control & Automation: Industrial Robot Programming (FANUC KAREL, Yamaha Robot Language), PLCs, Kinematics, Motor Control, ROS, OpenCV, MediaPipe

Electronics & Hardware: Circuit Design, PCB Design and Fabrication (KiCad), Signal Conditioning, Power Systems

Embedded Systems & Sensing: Arduino, Raspberry Pi, Microcontrollers, Sensor Integration

Programming: Python, C++, C, MATLAB

Mechanical Design & Prototyping: Fusion 360, Siemens NX & Teamcenter, 3D Printing, Laser Cutting, Hand and Power Tools

Manufacturing & Process Optimization: Lean Manufacturing (Kaizen, 5S), Value Stream Mapping, Cycle Time Analysis, Cost-Benefit Evaluation